



Report 1: Real Data Analysis

Code ▾

1 Social Media and Mental Health (Braghieri et al., 2022)

This report analyses a clinically validated survey dataset of US college students using PHQ-9 (depression) and GAD-7 (anxiety) screening instruments.

Dataset: Social Media and Mental Health
Source: Braghieri, Levy & Makarin (2022) via ICPSR
Records: 579 students across 48 US states
Variables: 118 columns including PHQ-9, GAD-7, ACHA health indicators, and demographic variables

2 1. Data Loading & Quality Checks

► Code

Libraries loaded successfully

Table styling functions loaded ✓

► Code

Dataset loaded: 579 rows × 118 columns

Dataset Preview — First 3 Rows (key columns)

	Birth Year	State	General Health	Sex	Full Time	International	PHQ-9 Score	GAD-7 Score
0	2000	MD	Very Good	Female	Yes	No	6	0
1	2001	SC	Good	Female	Yes	No	19	4
2	1999	NJ	Good	Female	Yes	No	10	9

3 2. Dataset Overview

Before any cleaning is applied, the dataset is inspected to understand its structure, identify data quality issues, and determine what preprocessing will be required.

3.1 2.1 Structure & Data Types

**df.shape**

579 rows / 118 cols

**isnull()**

3,396 missing values

**duplicated()**

0 duplicate rows

**dtypes**

49 numeric columns
69 object columns
0 boolean columns

3.2 2.2 Missing Values

► Code

Dataset: 579 rows × 118 columns

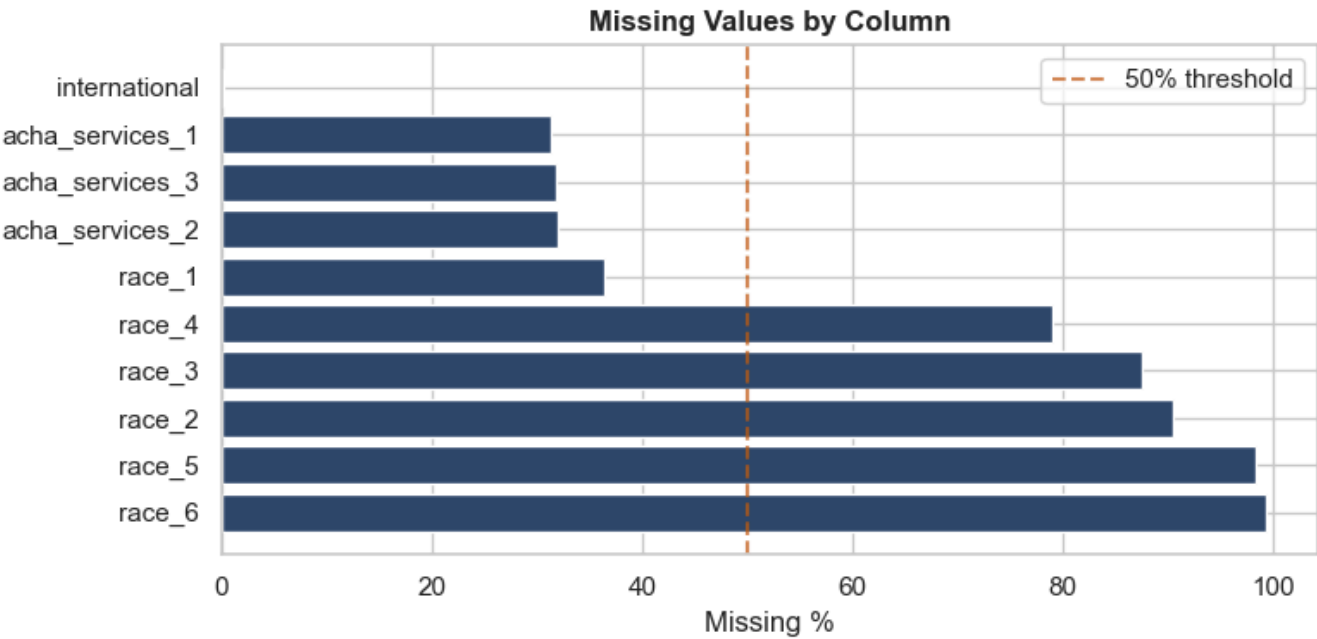
Total columns with missing values: 10

Total missing values: 3396

Missing Values by Column

	Missing Count	Missing %
race_6	575	99.31
race_5	570	98.45
race_2	524	90.50
race_3	507	87.56
race_4	458	79.10
race_1	211	36.44
acha_services_2	185	31.95
acha_services_3	184	31.78
acha_services_1	181	31.26

	Missing Count	Missing %
international	1	0.17



► Code

Dataset: 579 rows × 118 columns

Data type breakdown:

object 69
int64 49

Columns with Missing Values – Data Type & Null Summary

	Data Type	Non-Null Count	Null Count	Null %
race_6	object	4	575	99.31
race_5	object	9	570	98.45
race_2	object	55	524	90.50
race_3	object	72	507	87.56
race_4	object	121	458	79.10
race_1	object	368	211	36.44
acha_services_2	object	394	185	31.95
acha_services_3	object	395	184	31.78
acha_services_1	object	398	181	31.26
international	object	578	1	0.17

3.3 2.2 Interpreting the Missing Values

The table above reveals that not all missing values have the same cause. Two distinct patterns are identified:

Pattern 1 — Structural skip pattern (`acha_services_1/2/3`)

These columns ask whether a student was diagnosed, is in therapy, or is on medication — but only if they answered *Yes* to having depression (`acha_depression`). Students who answered *No* were never asked these questions. Their NaN values are **not missing data** — they are legitimately *not applicable*. Standard imputation would introduce false information here.

Pattern 2 — Multi-select wide format (`race_1` to `race_6`)

Students selected all racial identities that applied. Each `race_N` column contains the label if selected, or NaN if not. A NaN here means “*did not select this option*” — not “*did not answer*”. These columns require restructuring,

3.4 2.3 What We Will Fix

Based on the inspection above, the following issues are addressed in the preprocessing pipeline:

2.3 What We Will Fix — Preprocessing Plan

	Issue	Column(s)	Action
0	Rogue whitespace in column name	phq9_6 NUM	Strip whitespace from all column names
1	Skip-pattern NaNs	acha_services_1/2/3	Flag as 'not_applicable' — not true missing data
2	Wide-format race columns	race_1 to race_6	Restructure into single 'race' column
3	Age not directly available	year_1 (birth year)	Derive age column (2022 – birth year)
4	Missing value imputation	acha_services_1/2/3	Compare mode vs probability vs domain-informed

4 3. Data Preprocessing & Sorting

All preprocessing is applied to `df_clean` — the working copy created before this section. The original dataset `df` remains unmodified throughout.

► Code

Working copy created: 579 rows × 118 columns
Original dataset preserved: 579 rows × 118 columns
Are they the same object in memory? False
Original also saved to: data/processed/mental_health_ORIGINAL_BACKUP.csv

4.1 3.1 Fixing Column Name Error

During dataset inspection, two issues were identified in column naming:

1. **Rogue whitespace** — the column `phq9_6 NUM` contains a space before `NUM`, which would cause `KeyError` failures in any downstream analysis referencing this column by name.
2. **Inconsistent naming convention** — all other PHQ-9 numeric columns follow the pattern `phq9_XNUM` (no space), making `phq9_6 NUM` an outlier that breaks programmatic column selection.

Both issues are resolved by: - Stripping all leading/trailing whitespace from all column names - Explicitly renaming `phq9_6 NUM` → `phq9_6NUM` to match the consistent naming convention of all other PHQ-9 item columns

► Code

```
Columns with spaces in name (before fix): ['phq9_6 NUM']
Columns with spaces in name (after fix): []
```

✓ All column names cleaned successfully

4.2 3.2 Handling Skip-Pattern Missing Values

Not all missing values represent absent data. In this dataset, columns `acha_services_1` (Diagnosed in last 12 months), `acha_services_2` (Currently in therapy), and `acha_services_3` (Currently on medication) follow a **skip pattern**: they were only presented to students who answered *Yes* to having been diagnosed with depression (`acha_depression`). Students who answered *No* were never asked these questions; therefore, their NaN values are legitimately *not applicable*, not missing.

Three approaches are compared: - **Mode imputation**: fills NaN with the most frequent value - **Probability-based imputation**: fills NaN proportionally to the observed distribution - **Domain-informed flagging**: replaces NaN with `not_applicable` where `acha_depression == 'No'`

Standard imputation (modes 1 and 2) would introduce false information by treating skipped questions as unanswered ones. The domain-informed approach is the analytically correct method for this data structure.

► Code

```
Total records: 579
acha_depression = No: 378 (65.3%)
acha_depression = Yes: 201 (34.7%)
acha_depression = NaN: 0
```

Diagnosed in last 12 months (`acha_services_1`)

```
Total NaN: 181
NaN where depression=No: 181 (skip pattern)
NaN where depression=Yes: 0 (genuine missing)
```

Currently in therapy (`acha_services_2`)

```
Total NaN: 185
NaN where depression=No: 185 (skip pattern)
NaN where depression=Yes: 0 (genuine missing)
```

Currently on medication (`acha_services_3`)

Total NaN: 184

NaN where depression=No: 184 (skip pattern)

NaN where depression=Yes: 0 (genuine missing)

No imputation was required. All NaN values were structurally explained by the survey skip pattern and replaced with 'not_applicable' to preserve data integrity.

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Skip-Pattern Fix Summary

	Yes	No	not_applicable	NaN remaining	Total
Column					
Diagnosed in last 12 months (acha_services_1)	62	336	181	0	579
Currently in therapy (acha_services_2)	83	311	185	0	579
Currently on medication (acha_services_3)	122	273	184	0	579

4.3 3.3 Restructuring Race Columns (Wide → Long Format)

The dataset stores racial identity as six separate columns (`race_1` to `race_6`), each representing one category in a select-all-that-apply question. A value indicates the student selected that option; NaN indicates they did not.

This approach preserves multi-select responses while making the column analytically usable for grouping and visualisation.

Before — Wide Format (one column per category)

Student	race_1	race_2	race_3	race_4	race_5	race_6
A	White-not Hispanic	NaN	NaN	NaN	NaN	NaN
B	NaN	Black-not Hispanic	NaN	NaN	NaN	NaN
C	White-not Hispanic	NaN	Hispanic or Latino	NaN	NaN	NaN

After — Single Race Column

Student	Race
A	White-not Hispanic
B	Black-not Hispanic
C	White-not Hispanic, Hispanic or Latino

► Code

Race Distribution After Restructuring

Race	Count	%
White - not Hispanic (includes Middle Eastern)	327	56.5
Asian or Pacific Islander	103	17.8
Hispanic or Latino	61	10.5
Multiple	45	7.8
Black - not Hispanic	40	6.9
Other	3	0.5

4.4 3.4 Deriving Age from Birth Year

The dataset does not contain an age column directly. Instead, `year_1` records each student’s birth year (range: 1971–2003). Age is derived by subtracting the birth year from the survey year (2022).

Before: `year_1 = 1995` → **After:** `age = 27`

This derived variable is essential for demographic analysis and age-group comparisons in the visualisation section.

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Before & After — Birth Year to Derived Age

	Birth Year (before)	Derived Age (after)
Student 1	2000	22
Student 2	2001	21
Student 3	1999	23
Student 4	1998	24
Student 5	2000	22

Age Summary Statistics

	count	mean	std	min	25%	50%	75%	max
Age	579.0	23.6	5.4	19.0	21.0	22.0	24.0	51.0

Age Group Distribution

	Age Group	Count	%
0	19-21	239	41.3
1	22-24	217	37.5
2	25-27	39	6.7
3	28-30	18	3.1
4	30+	66	11.4

4.5 3.5 Sorting Algorithm — Merge Sort on PHQ-9 Score

To demonstrate algorithmic thinking, a merge sort algorithm is applied to order students by their PHQ-9 composite depression score (`phq9_COMP`).

Merge sort was selected for its efficiency and divide-and-conquer approach — it splits the dataset recursively, sorts each half, then merges them back in order. Unlike Python's built-in `sort()`, implementing merge sort explicitly demonstrates understanding of algorithmic complexity ($O(n \log n)$).

► Code

Top 10 — Lowest PHQ-9 Depression Scores (Minimal/No Depression)

Birth Year	Sex	PHQ-9 Score	Depression Severity	GAD-7 Score	Anxiety Severity
2000	Male	0	Minimal depression	6	mild anxiety
2000	Male	0	Minimal depression	13	moderate anxiety
1995	Female	0	Minimal depression	15	severe anxiety
1991	Male	0	Minimal depression	7	mild anxiety

Birth Year	Sex	PHQ-9 Score	Depression Severity	GAD-7 Score	Anxiety Severity
1994	Male	0	Minimal depression	8	mild anxiety
2001	Male	0	Minimal depression	0	minimal anxiety
1994	Female	0	Minimal depression	13	moderate anxiety
1997	Female	0	Minimal depression	1	minimal anxiety
2001	Female	0	Minimal depression	16	severe anxiety
2003	Male	0	Minimal depression	1	minimal anxiety

Top 10 — Highest PHQ-9 Depression Scores (Most Severe)

Birth Year	Sex	PHQ-9 Score	Depression Severity	GAD-7 Score	Anxiety Severity
1999	Female	27	Severe depression	14	moderate anxiety
1990	Male	27	Severe depression	21	severe anxiety
2000	Male	27	Severe depression	2	minimal anxiety
2001	Female	26	Severe depression	7	mild anxiety
2001	Male	26	Severe depression	13	moderate anxiety
2001	Female	25	Severe depression	4	minimal anxiety
2003	Female	25	Severe depression	8	mild anxiety
2002	Female	25	Severe depression	8	mild anxiety
2001	Female	25	Severe depression	9	mild anxiety

Birth Year	Sex	PHQ-9 Score	Depression Severity	GAD-7 Score	Anxiety Severity
1994	Female	25	Severe depression	14	moderate anxiety

Merge Sort – PHQ-9 Summary Statistics

Metric	Value
Total scores sorted	579.0
Minimum score	0.0
Maximum score	27.0
Median score	9.0
Mean score	9.4

► Code

Top 10 – Highest PHQ-9 Depression Scores (Most Severe)

Birth Year	Sex	PHQ-9 Score	Depression Severity	GAD-7 Score	Anxiety Severity
1990	Male	27	Severe depression	21	severe anxiety
1999	Female	27	Severe depression	14	moderate anxiety
2000	Male	27	Severe depression	2	minimal anxiety
2001	Male	26	Severe depression	13	moderate anxiety
2001	Female	26	Severe depression	7	mild anxiety
1994	Female	25	Severe depression	14	moderate anxiety
2001	Female	25	Severe depression	10	moderate anxiety
2001	Female	25	Severe depression	9	mild anxiety

Birth Year	Sex	PHQ-9 Score	Depression Severity	GAD-7 Score	Anxiety Severity
2002	Female	25	Severe depression	8	mild anxiety
2003	Female	25	Severe depression	8	mild anxiety

► Code

Depression Diagnosis — Population Breakdown

Group	Count	%	Mean PHQ-9	Mean GAD-7
Diagnosed with depression	201	34.7	9.7	10.2
Not diagnosed	378	65.3	9.2	7.3
Total	579	100.0	9.4	8.3

4.6 Key Insight — Diagnosed vs Non-Diagnosed

The table above reveals a striking finding:

Students **diagnosed with depression** (34.7%) have a mean PHQ-9 of **9.7** and mean GAD-7 of **10.2** — sitting at the *moderate* threshold for both depression and anxiety.

Students **not diagnosed** (65.3%) have a mean PHQ-9 of **9.2** and mean GAD-7 of **7.3** — only marginally lower on depression, but notably lower on anxiety.

The difference in PHQ-9 between the two groups is just **0.5 points** — clinically negligible. This suggests that a significant proportion of students without a formal depression diagnosis are experiencing symptom levels comparable to those who have been diagnosed.

This points to a likely **underdiagnosis problem** — students are experiencing moderate depression symptoms without having sought or received a formal diagnosis. The anxiety gap (10.2 vs 7.3) is more pronounced, suggesting that anxiety may be a stronger differentiator between diagnosed and undiagnosed students than depression severity alone.

💡 **This finding will be revisited in the conclusions section.**

4.7 3.6 Missing Value Imputation: Method Comparison

During dataset inspection, [acha_services_1](#) (*diagnosed in last 12 months*), [acha_services_2](#) (*currently in therapy*), and [acha_services_3](#) (*currently on medication*) appeared to have ~31% missing values each.

A less careful analyst might have applied standard imputation immediately. This section demonstrates what would have happened: and why it would have been **analytically wrong**.

By identifying the **skip pattern** in Section 3.2: that these questions were only asked to students who answered *Yes* to having depression — we avoided introducing false data into the dataset.

The three methods below are compared against the original distribution to show how each one distorts the data differently:

Method	What it does	Problem
Mode imputation	Fills all NaN with No (most frequent answer)	Inflates No responses: treats skipped questions as answered
Probability-based	Fills NaN proportionally to Yes/No distribution	Still treats skipped questions as answered: just more subtly
Domain-informed 	Flags NaN as not_applicable where acha_depression = No	Correctly reflects survey structure: no false data introduced

 The correct approach is not always the most obvious one. Understanding *why* data is missing is more important than simply filling it in.

► Code

Imputation Comparison — Diagnosed in last 12 months (acha_services_1)

Value	Original (non-NaN)	Mode Imputation	Probability-Based	Domain-Informed 
Yes	15.6%	10.7%	10.7%	10.7%
No	84.4%	58.0%	58.0%	58.0%
not_applicable	—	—	—	31.3%
NaN remaining values	181	0	0	0
✓ Method chosen?	—	 No	 No	 Yes — correctly reflects skip pattern

Imputation Comparison — Currently in therapy (acha_services_2)

Value	Original (non- NaN)	Mode Imputation	Probability- Based	Domain- Informed ✓
Yes	21.1%	14.3%	14.3%	14.3%
No	78.9%	53.7%	53.7%	53.7%
not_applicable	—	—	—	32.0%
NaN remaining	185 values	0	0	0
✓ Method chosen?	—	✗ No	✗ No	✓ Yes — correctly reflects skip pattern

Imputation Comparison — Currently on medication (acha_services_3)

Value	Original (non- NaN)	Mode Imputation	Probability- Based	Domain- Informed ✓
Yes	30.9%	21.1%	21.1%	21.1%
No	69.1%	47.2%	47.2%	47.2%
not_applicable	—	—	—	31.8%
NaN remaining	184 values	0	0	0
✓ Method chosen?	—	✗ No	✗ No	✓ Yes — correctly reflects skip pattern

4.8 3.7 Outlier Detection Using IQR

Outliers are identified using the **Interquartile Range (IQR)** method across the key numeric variables. The IQR is the range between the 25th and 75th percentiles: values beyond $1.5 \times \text{IQR}$ above or below this range are flagged as outliers.

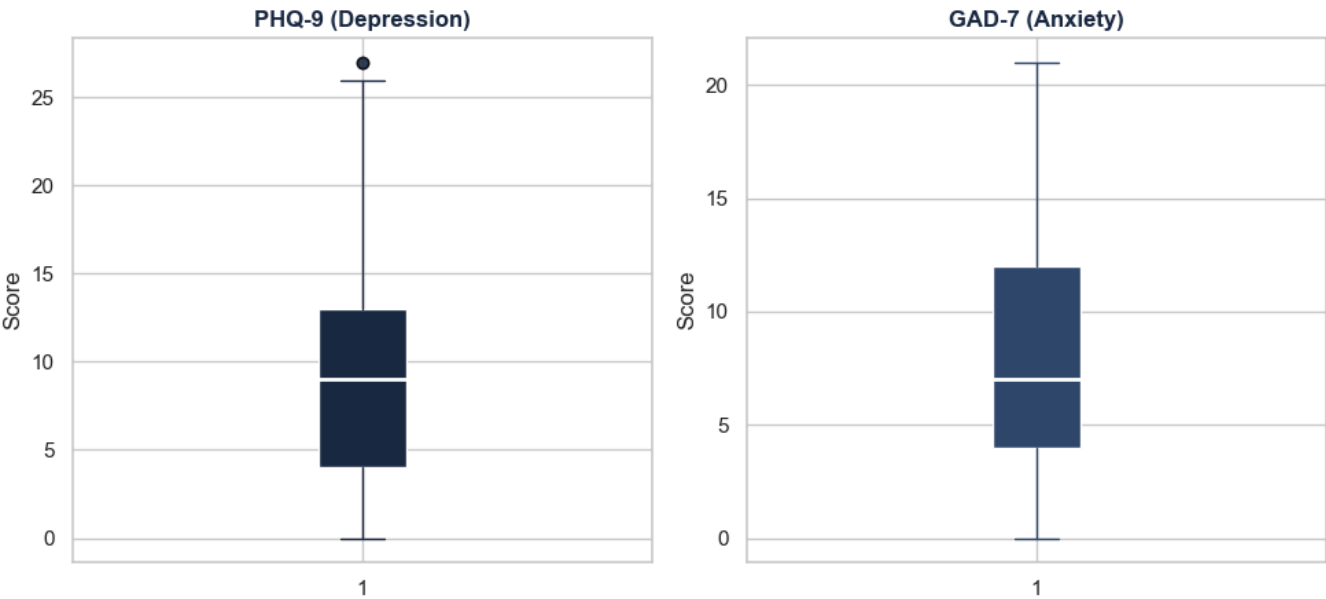
Outliers are retained in the dataset as they likely represent genuine extreme cases rather than data entry errors: a student scoring 27 on PHQ-9 is a real clinical presentation, not a mistake.

► Code

Outlier Detection — IQR Method

Variable	Q1	Q3	IQR	Lower Bound	Upper Bound	Outliers Detected
PHQ-9 (Depression)	4.0	13.0	9.0	-9.5	26.5	3
GAD-7 (Anxiety)	4.0	12.0	8.0	-8.0	24.0	0
Age	21.0	24.0	3.0	16.5	28.5	72

Outlier Detection — Key Clinical Variables



4.9 3.7.1 Outlier Decision — Retain

The IQR method identified **3 outliers in PHQ-9** (scores above 26.5) and **0 outliers in GAD-7**.

In this dataset, outliers are **retained** for the following reasons:

Reason	Explanation
Clinical validity	A PHQ-9 score of 27 is the maximum possible: it represents a real student experiencing severe depression, not a data entry error
Instrument design	PHQ-9 and GAD-7 are bounded scales (0–27 and 0–21): extreme scores are expected and meaningful
Research integrity	Removing high scorers would systematically bias results toward underestimating depression severity

Reason	Explanation
Small number	Only 3 outliers in 579 records (0.5%): insufficient to distort analysis

Removing outliers from clinical mental health data risks silencing the most vulnerable individuals in the dataset: the opposite of what good research should do.

4.10 3.8 Converting Frequency Columns to Numeric

The `acha_12months_times` columns record how often students experienced specific mental health events in the past 12 months. Values are stored as text frequency labels rather than numbers, making them unusable for correlation analysis or any numeric computation.

A frequency mapping is applied to convert these to ordinal numeric values (required for further data visualisation):

Text Value	Numeric
Never	0
1-2 times	1
3-4 times	2
5-6 times	3
7-8 times	4
9-10 times	5
11 or more times	6

► Code

Before — Text Frequency Values (showing 3 of 7 columns)

acha_12months_times_1	acha_12months_times_2	acha_12months_times_3
Never	11 or more times	5-6 times
1-2 times	5-6 times	7-8 times
7-8 times	11 or more times	11 or more times

After — Converted to Numeric (showing 3 of 7 columns)

acha_12months_times_1	acha_12months_times_2	acha_12months_times_3
0	6	3
1	3	4
4	6	6

Frequency Mapping — Text to Numeric

Text Value (before)	Numeric (after)	Severity
Never	0	None
1-2 times	1	Very Low
3-4 times	2	Low
5-6 times	3	Moderate
7-8 times	4	High
9-10 times	5	Very High
11 or more times	6	Extreme

Total columns converted: 7 ✓

5 4. Functional Programming

Functional programming techniques: `map()`, `filter()`, and `reduce()` are applied to transform, refine, and aggregate the dataset.

These approaches support efficient data processing and demonstrate understanding of functional programming concepts in data analysis. Each technique is applied to clinically meaningful variables, producing results that directly inform the analysis in Section 5.

Function	Purpose	Applied To
<code>map()</code>	Transform each element	PHQ-9 and GAD-7 scores → severity categories
<code>filter()</code>	Select elements meeting a condition	Student subgroups by clinical severity
<code>reduce()</code>	Aggregate elements into a single result	Total and mean depression/anxiety scores

5.1 4.1 Data Transformation Using `map()`

`map()` applies a function to every element in a sequence efficiently and without explicit loops. Here it is used to translate raw PHQ-9 and GAD-7 numeric scores into clinically meaningful severity categories: making the

data immediately interpretable without modifying the original composite score columns.

Scale	Score Range	Categories
PHQ-9 (Depression)	0–27	Minimal / Mild / Moderate / Moderately Severe / Severe
GAD-7 (Anxiety)	0–21	Minimal / Mild / Moderate / Severe

Two new columns are created: `phq9_category` and `gad7_category`.

► Code

PHQ-9 Depression Severity — `map()` Applied

Category	Count	%
Severe	38	6.6
Moderately Severe	78	13.5
Moderate	147	25.4
Mild	170	29.4
Minimal	146	25.2

GAD-7 Anxiety Severity — `map()` Applied

Category	Count	%
Severe	93	16.1
Moderate	123	21.2
Mild	191	33.0
Minimal	172	29.7

5.2 4.2 Data Selection Using `filter()`

`filter()` selects elements from a sequence that satisfy a given condition, returning only the subset that meets the criteria. Here it is applied to identify clinically significant student subgroups based on PHQ-9 and GAD-7 thresholds.

Three subgroups are isolated:

Subgroup	Threshold	Clinical Meaning
Moderate+ depression	PHQ-9 $\geq$ 10	Requires clinical attention
Severe anxiety	GAD-7 $\geq$ 15	Significant functional impairment

Subgroup	Threshold	Clinical Meaning
Both conditions	PHQ-9 ≥ 10 AND GAD-7 ≥ 15	Comorbid — highest risk group

This approach demonstrates how `filter()` supports targeted analysis without modifying the original dataset.

► Code

`filter()` — Clinical Subgroup Selection

Subgroup	Count	% of total
Moderate+ depression (PHQ-9 ≥ 10)	263	45.4
Severe anxiety (GAD-7 ≥ 15)	93	16.1
Both conditions combined	45	7.8

5.3 4.3 Data Aggregation Using `reduce()`

`reduce()` iteratively applies a function to a sequence, accumulating the result into a single value. It is imported from Python’s `functools` module and is particularly useful for aggregating large sequences without explicit loops.

Here `reduce()` is used to calculate total and cumulative scores across the dataset — demonstrating how functional aggregation works under the hood compared to pandas `.sum()`.

► Code

`reduce()` — Aggregated PHQ-9 and GAD-7 Statistics

Metric	PHQ-9 (Depression)	GAD-7 (Anxiety)
Total score (all students)	5427.0	4820.0
Mean score per student	9.4	8.3
Maximum score	27.0	21.0
Minimum score	0.0	0.0
Students assessed	579.0	579.0

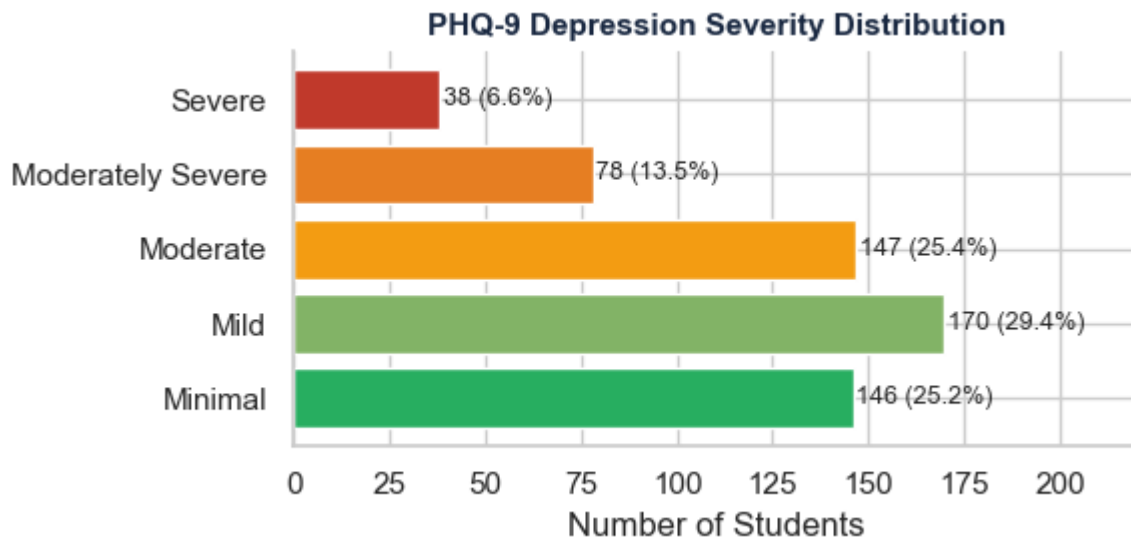
6 5. Data Visualisation

This section presents key visual insights derived from the dataset. Visualisations are used to support the interpretation of mental health patterns across the student population, with a focus on depression and anxiety severity, demographic differences, comorbidity, and treatment-seeking behaviour.

All charts use a consistent navy colour palette to maintain visual coherence throughout the report.

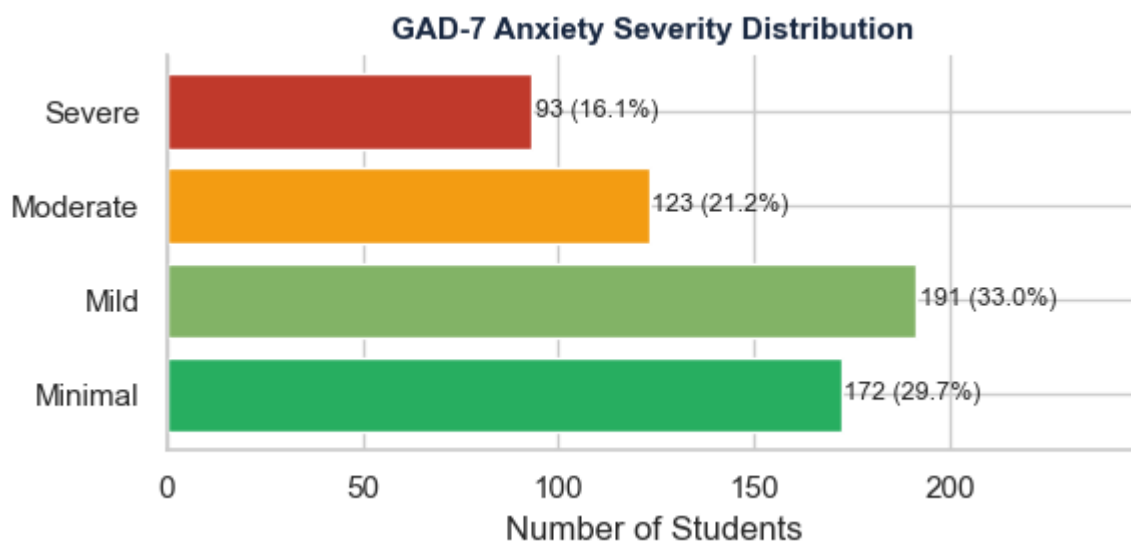
6.1 5.1 PHQ-9 Depression Severity Distribution

► Code



6.2 5.2 GAD-7 Anxiety Severity Distribution

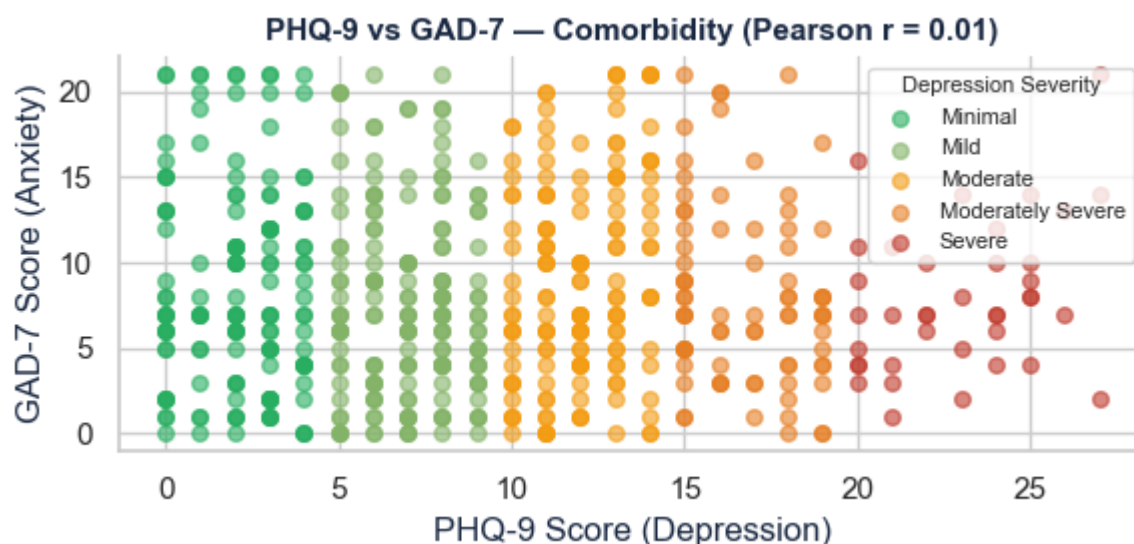
► Code



6.3 5.3 Depression vs Anxiety — Comorbidity Scatter Plot

Each point represents one student. The scatter plot reveals the relationship between PHQ-9 (depression) and GAD-7 (anxiety) scores: a strong positive correlation would confirm high comorbidity between the two conditions.

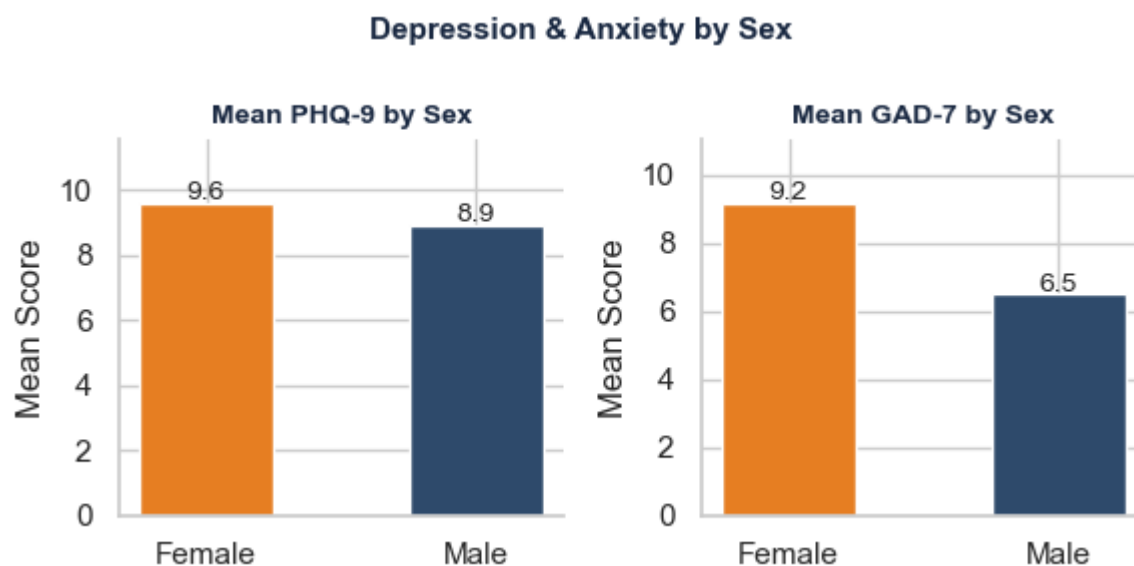
► Code



6.4 5.4 Depression and Anxiety by Sex

Comparing PHQ-9 and GAD-7 mean scores between male and female students reveals whether depression and anxiety severity differs by sex: a clinically relevant demographic comparison.

► Code



Key Finding:

Female students score higher than male students on > both PHQ-9 (depression) and GAD-7 (anxiety): consistent with well-established clinical literature showing higher rates of internalising disorders among women. This pattern is not unique to this dataset: it reflects a systemic difference in how depression and anxiety manifest and are reported across sexes.

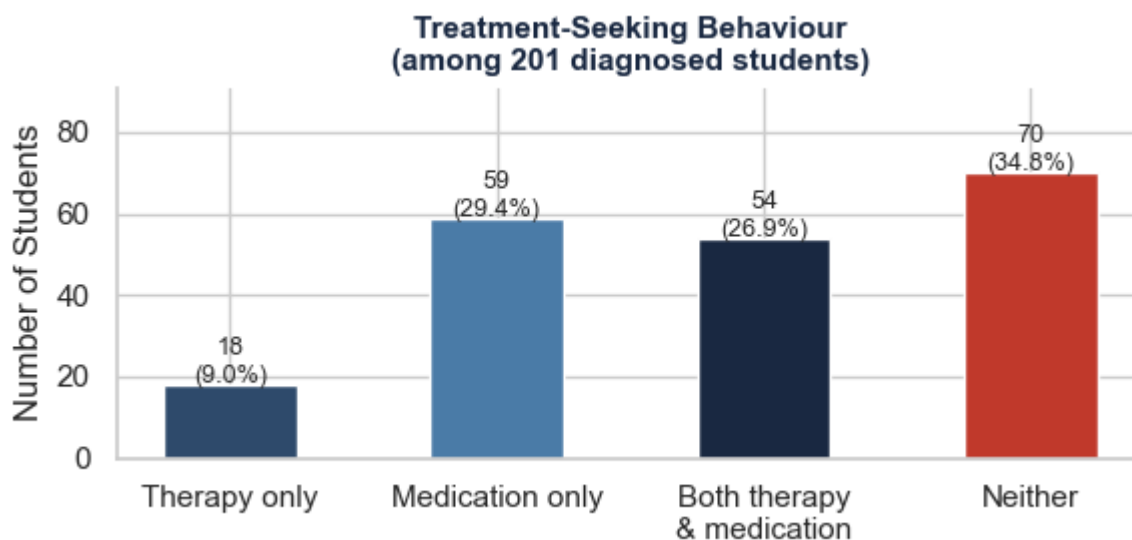
Importantly, higher scores do not necessarily mean women suffer more; they may also reflect greater willingness to acknowledge and report symptoms, while men may underreport due to social stigma

around mental health disclosure.

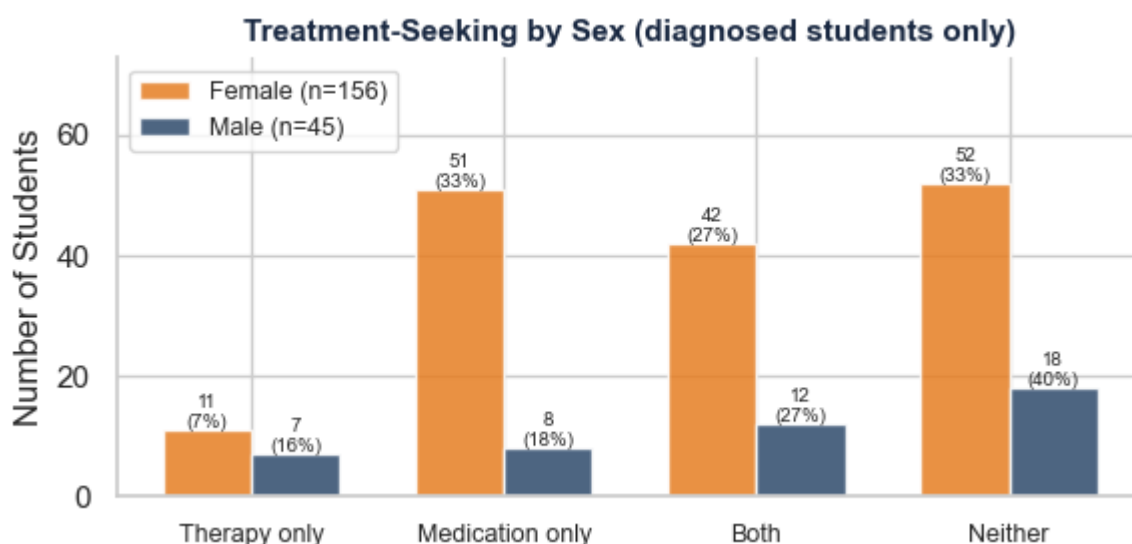
6.5 5.5 Treatment-Seeking Behaviour

Among students diagnosed with depression, this section examines what proportion are actively receiving treatment: either through therapy or medication. Low treatment rates relative to diagnosis rates would suggest a significant gap between clinical need and help-seeking behaviour.

► Code



► Code



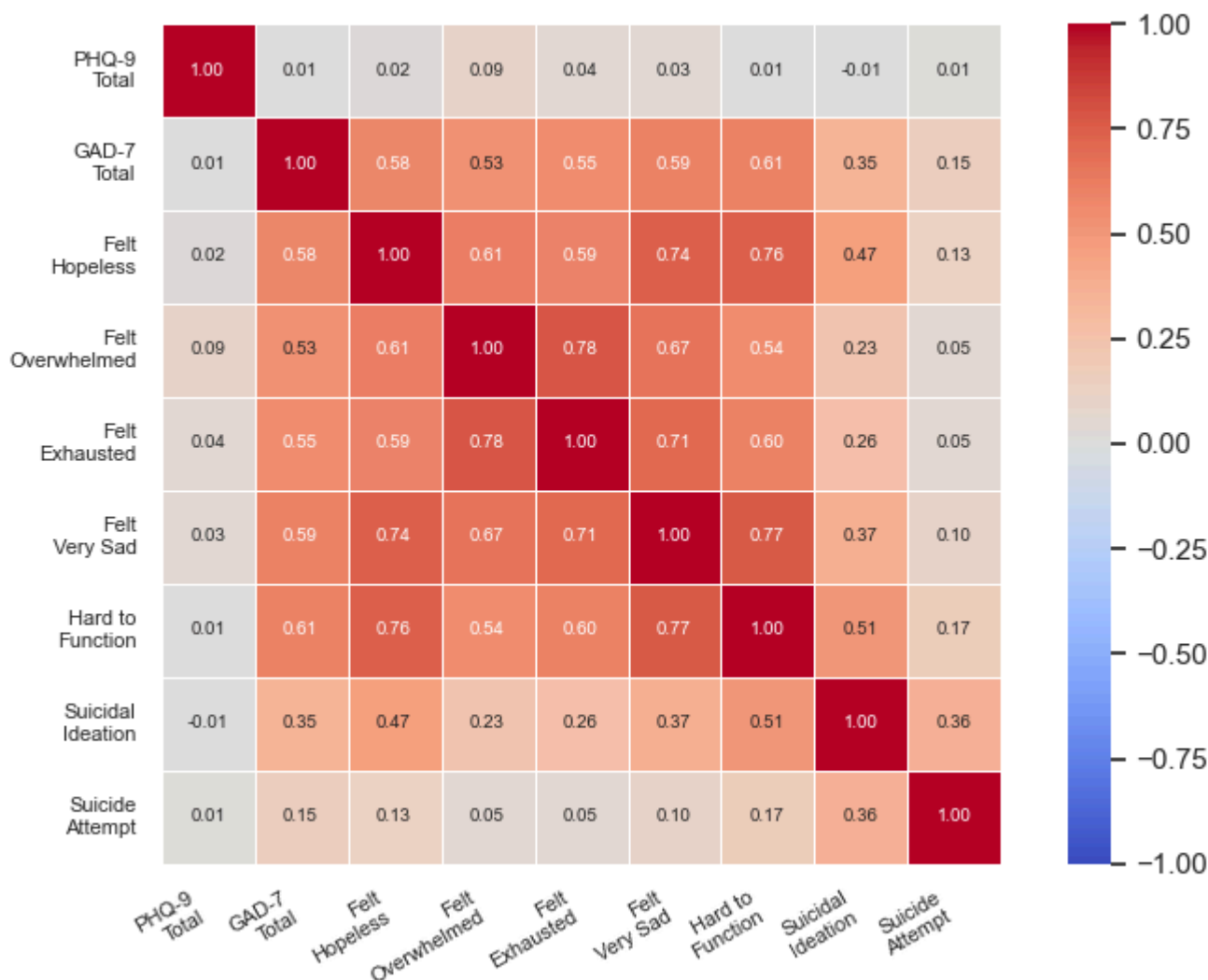
Key Finding: Approximately one third of female students diagnosed with depression are receiving neither therapy nor medication. Combined with the earlier finding that undiagnosed students show similar PHQ-9 scores to diagnosed ones, this suggests a substantial **treatment gap** — where clinical need significantly exceeds active help-seeking behaviour, particularly among women who have already crossed the threshold of formal diagnosis.

6.6 5.6 Correlation Heatmap

A correlation heatmap reveals the strength and direction of relationships between key numeric variables. Values close to 1 indicate a strong positive correlation, values close to -1 indicate a strong negative correlation, and values close to 0 indicate little relationship.

► Code

**Heatmap 1 — Initial Analysis: All Variables Combined
(reveals timeframe mismatch problem)**



6.7 Correlation Analysis — A Story of Discovery

Initial analysis combined PHQ-9, GAD-7, and the 12-month mental health event variables into a single heatmap (Heatmap 1). The result was puzzling — PHQ-9 (depression) showed near-zero correlation with almost everything, while GAD-7 (anxiety) showed meaningful correlations with the 12-month variables.

This prompted a closer look at the measurement timeframes:

- **PHQ-9** measures symptom severity over the **last 2 weeks**
- **GAD-7** measures symptom severity over the **last 2 weeks**
- **acha_12months_times** measures event frequency over the **last 12 months**

The low correlations were not a data error — they were a **methodological signal**. Comparing variables measured over different timeframes produces artificially diluted correlations. A student may have felt hopeless frequently across the year but be experiencing a better fortnight right now.

This led to splitting the analysis into three separate heatmaps — one per instrument and timeframe — which produced far more meaningful results.

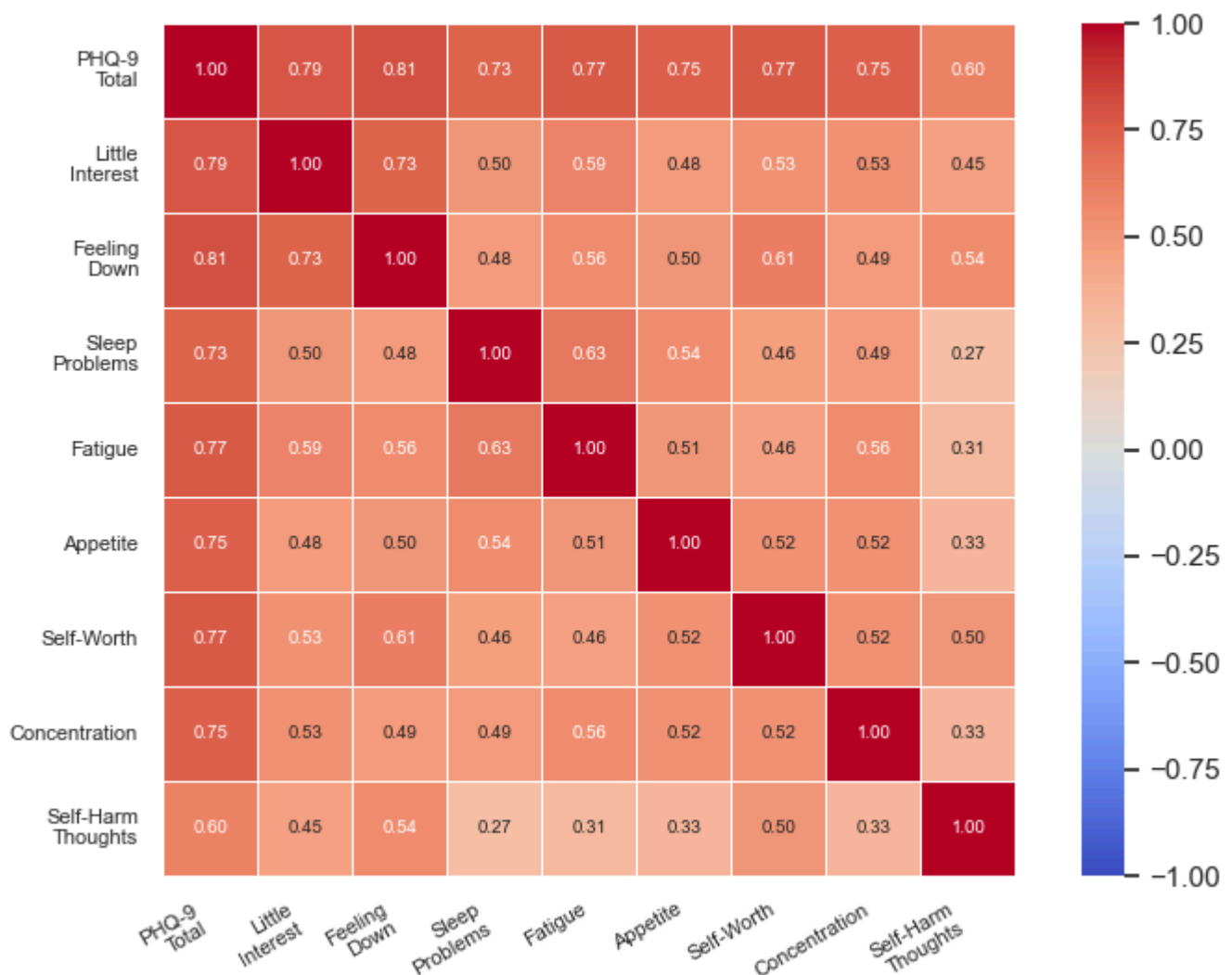
An unexpected finding emerged from this separation.

Even within the same 2-week timeframe, GAD-7 (anxiety) and PHQ-9 (depression) show near-zero correlation with each other ($r = 0.01$) — even at the individual symptom item level ($r = 0.00$ to 0.03). Both instruments are correctly scored and normally distributed — this is not a data error.

This suggests that in this population, depression and anxiety operate as largely **independent conditions**. A student experiencing severe depression is not necessarily experiencing anxiety — and vice versa. This challenges the common assumption that the two always co-occur, and has real implications for treatment — addressing one condition may not automatically address the other.

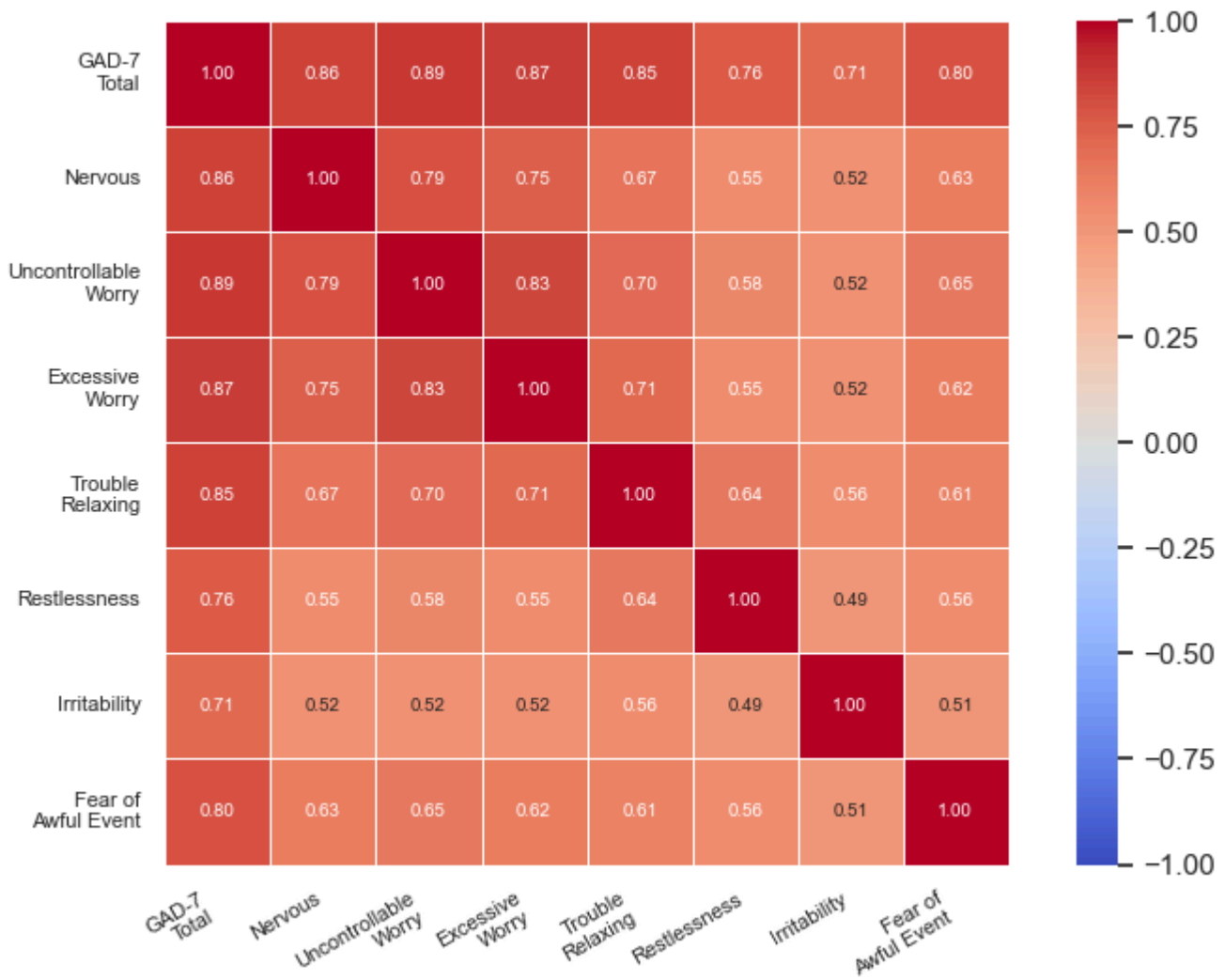
► Code

Heatmap 2 — PHQ-9 Depression Symptom Correlations (last 2 weeks)



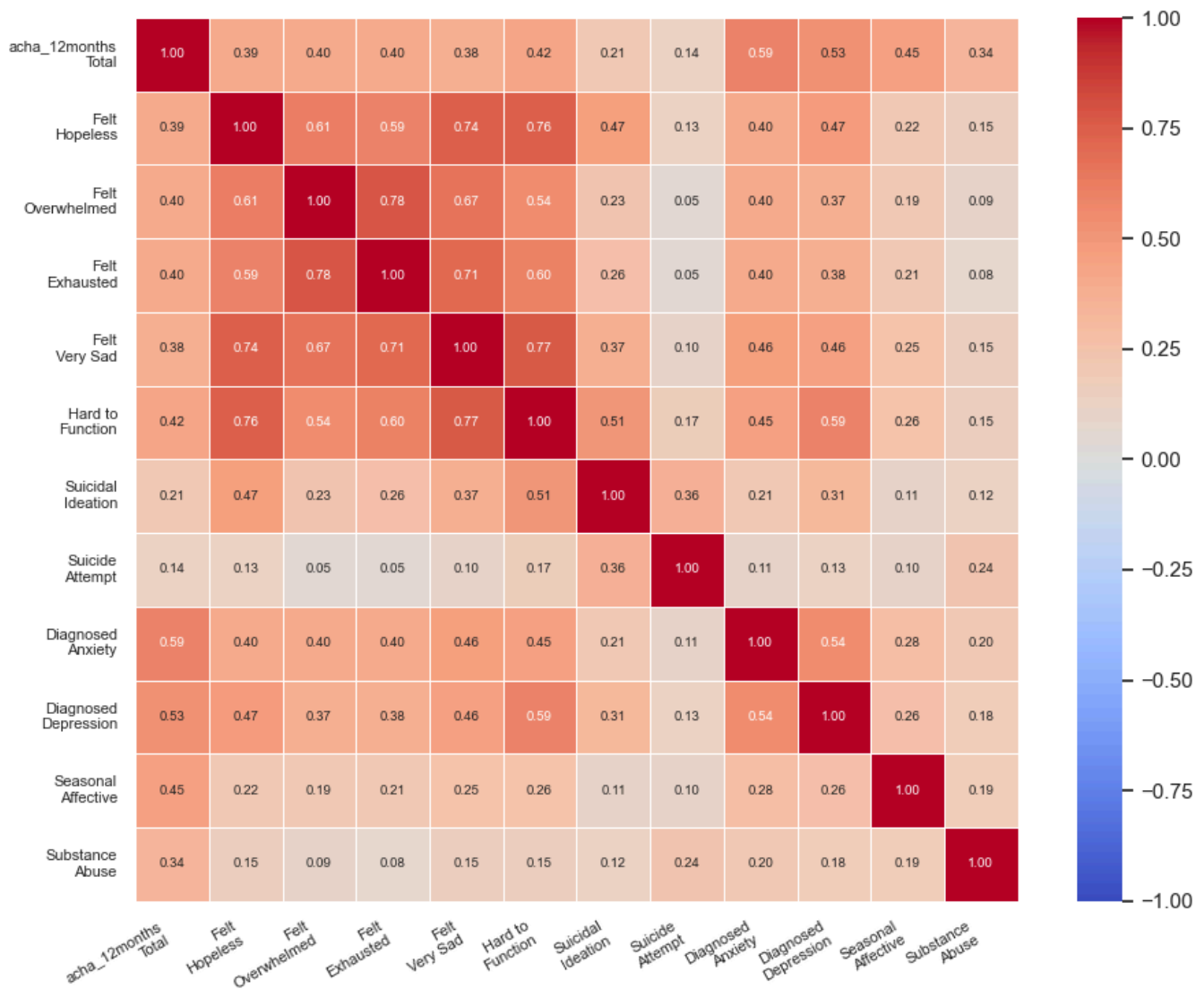
► Code

**Heatmap 3 — GAD-7 Anxiety Symptom Correlations
(last 2 weeks)**



► Code

**Heatmap 4 — 12-Month Mental Health Events & Diagnoses
(last 12 months)**



7 6. Conclusions

This report analysed the Social Media and Mental Health dataset (Braghieri et al., 2022) using a range of data wrangling and analytical techniques. The following key findings emerged:

7.1 6.1 Key Findings

Depression and anxiety are widespread. 74.8% of students score Mild or above on PHQ-9 and 70.3% on GAD-7 — moderate-to-severe mental health symptoms are the norm rather than the exception in this population.

Depression and anxiety are surprisingly independent. Despite both being highly prevalent, PHQ-9 and GAD-7 show near-zero correlation ($r = 0.01$) — even at the individual symptom item level. In this population, experiencing depression does not predict experiencing anxiety and vice versa.

Female students carry a heavier burden. Female students score higher than male students on both PHQ-9 and GAD-7 — consistent with clinical literature, though may also reflect greater willingness to acknowledge symptoms rather than a straightforward difference in prevalence.

Diagnosis does not guarantee treatment. Among 201 formally diagnosed students, approximately one third of females receive neither therapy nor medication — a significant treatment gap that persists even after the threshold of formal diagnosis has been crossed.

Underdiagnosis is likely. Undiagnosed students show PHQ-9 scores only marginally lower than diagnosed ones (9.2 vs 9.7) — a clinically negligible difference suggesting widespread unrecognised mental health need in this population.

Sustained distress clusters around hopelessness and functional impairment. The 12-month heatmap reveals that Felt Hopeless, Felt Very Sad, and Hard to Function form a tight correlation cluster ($r = 0.74\text{--}0.78$) — representing the emotional core of persistent mental health distress. Suicide Attempt sits largely isolated from this cluster ($r = 0.05\text{--}0.17$), consistent with clinical understanding that suicidal behaviour is not simply an extension of emotional distress frequency.

7.2 6.2 Limitations

- **Population scope:** The dataset covers US college students only — findings cannot be generalised to other age groups, non-student populations, or other countries including Australia.
- **Cross-sectional design:** The dataset captures a single point in time. No causal conclusions can be drawn — it is not possible to determine whether social media exposure caused depression, or whether depressed students use social media more.
- **Self-report bias:** Both PHQ-9 and GAD-7 rely on self-reported symptoms. Students may underreport (stigma) or overreport (seeking help) — particularly along sex lines, which may partly explain the observed sex differences.
- **Timeframe mismatch:** PHQ-9 and GAD-7 measure the last 2 weeks while `acha_12months` variables measure the last 12 months — direct comparison between these instruments is not appropriate, as demonstrated in the correlation analysis.
- **No social media variables:** Despite being attributed to a social media study, the dataset contains no direct social media usage variables — limiting conclusions about the relationship between digital behaviour and mental health outcomes.

7.3 6.3 Outlook and Future Research Directions

This analysis opens several avenues for further investigation:

- **Longitudinal tracking:** A follow-up study tracking the same students over multiple semesters would allow causal relationships to be established — particularly whether depression severity predicts future treatment seeking.
- **Combining with behavioural data:** Linking this clinical dataset with actual social media usage data (screen time, platform type, posting frequency) would allow direct testing of the social media — mental health relationship that this dataset implies but cannot prove.
- **Australian context:** The WA Emergency Department dataset analysed in Report 2 provides a complementary population-level view. Future research could examine whether the treatment gap observed here (diagnosed but untreated students) contributes to the ED mental health presentation burden observed in WA.

- **Sex-disaggregated treatment pathways:** Given the observed differences in treatment-seeking between male and female students, future research could examine whether different intervention strategies are needed by sex — particularly for male students who may be less likely to seek help despite similar symptom severity.
- **Independence of depression and anxiety:** The surprising finding that PHQ-9 and GAD-7 are near-uncorrelated in this population warrants replication in other datasets — if confirmed, it has significant implications for treatment protocols that currently assume high comorbidity.